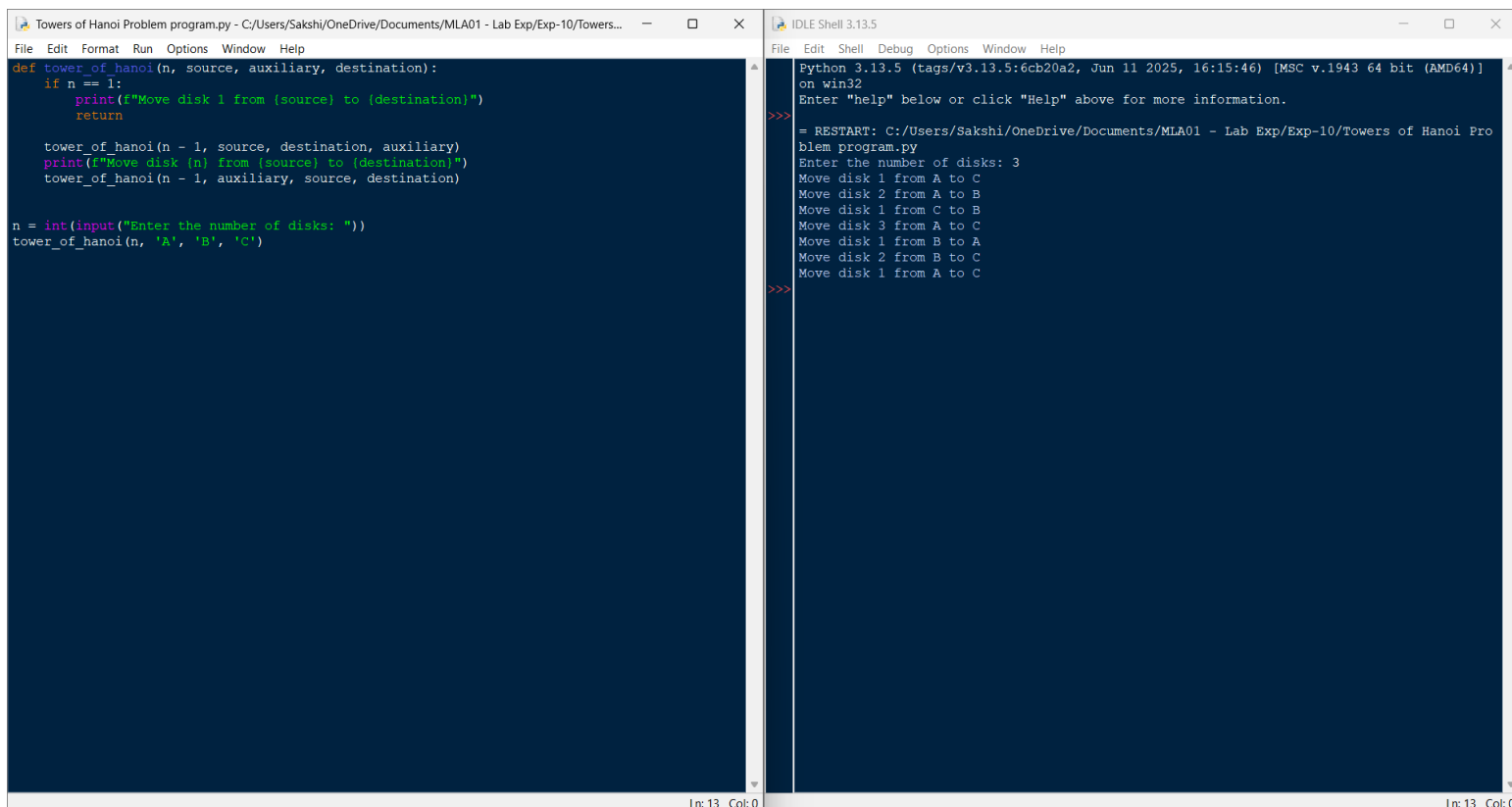


Exp:10 Towers of Hanoi Problem

Program:

```
def tower_of_hanoi(n, source, auxiliary, destination):  
  
    if n == 1:  
  
        print(f"Move disk 1 from {source} to {destination}")  
  
        return  
  
    tower_of_hanoi(n - 1, source, destination, auxiliary)  
  
    print(f"Move disk {n} from {source} to {destination}")  
  
    tower_of_hanoi(n - 1, auxiliary, source, destination)  
  
n = int(input("Enter the number of disks: "))  
  
tower_of_hanoi(n, 'A', 'B', 'C')
```

Output:



The screenshot displays a Python IDE with two windows. The left window, titled 'Towers of Hanoi Problem program.py', contains the source code for the Towers of Hanoi algorithm. The right window, titled 'IDLE Shell 3.13.5', shows the program's execution. The user has entered '3' for the number of disks, and the program has printed the sequence of moves required to solve the puzzle: Move disk 1 from A to C, Move disk 2 from A to B, Move disk 1 from C to B, Move disk 3 from A to C, Move disk 1 from B to A, Move disk 2 from B to C, and Move disk 1 from A to C.

```
Towers of Hanoi Problem program.py - C:/Users/Sakshi/OneDrive/Documents/MLA01 - Lab Exp/Exp-10/Towers...  
File Edit Format Run Options Window Help  
def tower_of_hanoi(n, source, auxiliary, destination):  
    if n == 1:  
        print(f"Move disk 1 from {source} to {destination}")  
        return  
  
    tower_of_hanoi(n - 1, source, destination, auxiliary)  
    print(f"Move disk {n} from {source} to {destination}")  
    tower_of_hanoi(n - 1, auxiliary, source, destination)  
  
n = int(input("Enter the number of disks: "))  
tower_of_hanoi(n, 'A', 'B', 'C')
```

```
IDLE Shell 3.13.5  
File Edit Shell Debug Options Window Help  
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]  
on win32  
Enter "help" below or click "Help" above for more information.  
>>>  
= RESTART: C:/Users/Sakshi/OneDrive/Documents/MLA01 - Lab Exp/Exp-10/Towers of Hanoi Pro  
blem program.py  
Enter the number of disks: 3  
Move disk 1 from A to C  
Move disk 2 from A to B  
Move disk 1 from C to B  
Move disk 3 from A to C  
Move disk 1 from B to A  
Move disk 2 from B to C  
Move disk 1 from A to C  
>>>
```